

MARADI, Niger

WMO: 610800

Lat: 13.503N

Lon: 7.127E

Elev: 378

StdP: 96.87

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.0	14.2	-11.0	1.5	28.5	-8.2	2.0	26.8	10.3	22.5	9.4	22.6	1.9	80	0.373

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	11.7	41.2	20.8	40.5	20.8	39.8	20.7	26.6	33.3	26.2	32.7	25.8	32.2	2.4	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.0	21.1	29.2	24.6	20.5	28.9	24.2	20.0	28.7	85.4	33.3	83.7	32.8	82.1	32.3	31.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.3	7.1	6.0	10.8	43.1	1.0	1.0	10.1	43.8	9.5	44.4	9.0	44.9	8.2	45.6		
			4.2	28.5	0.7	1.0	3.7	29.2	3.3	29.8	2.9	30.4	2.5	31.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.3	22.5	25.3	29.8	32.9	33.5	31.3	28.7	27.2	28.7	29.6	26.6	23.1
	DBStd	3.93	2.55	3.12	2.37	1.65	1.76	2.07	1.86	1.54	1.50	1.37	2.00	2.02
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	1	1	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6672	388	429	614	688	728	641	580	532	561	607	499	406
	CDD18.3	3631	131	196	356	438	469	391	322	274	311	349	249	147
	CDH23.3	48608	1469	2436	5204	7024	7394	5613	3877	2727	3613	4528	3102	1620
	CDH26.7	27871	638	1303	3287	4761	4964	3344	1824	1018	1709	2555	1762	705
Wind	WSAvg	2.4	3.2	2.9	2.6	2.1	2.3	2.5	2.4	1.9	1.6	1.7	2.3	3.2
Precipitation	PrecAvg	490	0	0	0	4	17	61	149	181	81	6	0	0
	PrecMax	730	0	0	6	38	65	156	270	342	203	74	5	5
	PrecMin	278	0	0	0	0	0	12	33	2	11	0	0	0
	PrecStd	131	0	0	1	8	16	33	56	64	46	13	1	1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.1	39.3	41.0	42.6	42.2	40.0	36.9	34.1	37.0	38.4	38.0	35.4
		MCWB	16.4	17.8	19.3	19.7	21.8	24.4	24.8	25.9	24.2	20.4	18.3	16.9
	2%	DB	33.9	37.2	40.1	41.7	41.2	38.8	35.4	33.0	35.5	37.7	37.0	33.8
		MCWB	15.4	17.1	18.6	19.8	22.2	24.3	24.5	25.7	25.0	20.5	18.2	16.2
	5%	DB	32.2	35.8	39.2	40.9	40.3	37.8	34.3	32.1	34.4	36.9	35.9	32.2
		MCWB	14.4	16.4	18.1	20.0	22.6	24.4	24.6	25.3	25.3	20.8	17.4	15.0
	10%	DB	30.3	33.9	38.0	39.9	39.4	36.7	33.2	31.1	33.2	35.9	34.7	30.8
		MCWB	13.8	15.4	17.4	19.7	22.8	24.3	24.5	25.0	25.3	21.1	16.8	14.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.1	21.0	23.3	25.5	26.6	26.7	26.6	26.7	27.2	26.4	21.6	17.9
		MCDB	33.7	34.0	35.8	37.0	36.3	35.7	32.8	32.3	33.3	33.5	33.4	33.8
	2%	WB	16.1	18.4	21.6	24.5	25.8	26.0	26.0	26.2	26.6	25.7	19.4	16.7
		MCDB	32.5	35.3	34.3	36.0	35.1	34.6	32.0	31.5	32.6	32.7	33.3	32.2
	5%	WB	14.9	16.9	19.7	23.8	25.3	25.6	25.5	25.8	26.2	25.2	18.3	15.8
		MCDB	31.2	33.9	37.3	35.0	34.8	33.9	31.4	30.7	32.0	31.8	33.5	31.2
	10%	WB	13.9	15.8	18.3	23.0	24.8	25.2	25.1	25.4	25.8	24.6	17.4	14.7
		MCDB	29.9	33.3	36.3	34.3	34.5	33.2	30.9	30.0	31.3	31.2	32.8	29.4
Mean Daily Temperature Range		MDBR	15.4	15.4	15.4	14.2	11.7	10.0	8.6	7.8	9.5	13.1	16.1	15.5
		MCDBR	17.6	17.1	16.8	15.2	13.0	11.1	9.9	8.9	11.0	14.5	16.9	17.3
		MCWBR	7.5	7.0	6.8	6.3	4.8	3.1	2.9	3.3	3.4	4.8	6.0	7.2
	5% WB	MCDDBR	17.2	16.4	15.4	12.4	11.3	10.3	8.9	8.5	10.0	11.5	16.3	17.0
		MCWBR	7.7	7.3	7.1	5.5	4.3	3.4	3.2	3.5	3.7	4.0	6.0	7.3
Clear-Sky Solar Irradiance	taub	0.579	0.640	0.693	0.725	0.708	0.681	0.623	0.595	0.598	0.634	0.552	0.541	
	taud	1.696	1.548	1.458	1.451	1.501	1.584	1.752	1.815	1.792	1.652	1.720	1.758	
	Ebn at Noon	723	700	678	655	655	664	707	737	737	696	739	741	
	Edh at Noon	235	283	316	317	297	269	229	218	223	251	227	216	
All-Sky Solar Radiation	RadAvg	5.57	6.08	6.44	6.64	6.48	6.25	5.99	5.76	6.25	6.16	5.80	5.44	
	RadStd	0.22	0.20	0.28	0.23	0.22	0.22	0.23	0.22	0.16	0.20	0.15	0.21	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page